View Reviews

Paper ID
686

Paper Title
Comparative Analysis of CNN-Based and Vision Transformer-Based Image Caption Generator Using Transformer Models

Track Name
Artificial Intelligence

Reviewer #2

Questions

1. Is the paper relevant to the scope of the conference?
The paper is relevant to conferences covering artificial intelligence, computer vision, deep learning, natural language proce

2. Technical quality: Provide comments based on the methodology, mathematical justification for proposed work, architecture/block diagram of proposed method, and experimental validation
The paper presents a systematic comparison of ResNet-50 and ViT-B/16 visual encoders using a common Transformer decoder. The methodology is reasonably detailed, covering pretrained encoders, vocabulary construction, training configuration, hyperparameter tuning, evaluation metrics, and computational efficiency. The use of Flickr30k and the reporting of BLEU-1 to BLEU-4, METEOR, ROUGE-L, and CIDEr provide a useful basis for comparing caption quality. The inclusion of parameter counts, training time, throughput, and inference latency is also valuable.

However, several aspects require further attention. The authors should clarify how the CNN's global feature vector and the ViT classification token provide comparable visual information to the shared decoder, since both approaches may differ in their ability to preserve spatial details. The experimental results should be strengthened through multiple independent training runs, variability estimates, and statistical significance testing because the observed differences in caption metrics are relatively small. A detailed comparison of generated captions, including examples of incorrect object descriptions and missed relationships, would improve the qualitative evaluation. The authors should also clarify the metric implementation, decoding configuration, and preprocessing consistency. Although a fixed seed and code repository are reported, complete software-version details and sufficient instructions for reproducing the reported results should be ensured. These improvements would strengthen the reliability and reproducibility of the experimental conclusions.

4. Novelty and Societal Impact: Provide comments on the originality of the proposed work and its potential societal, industrial, environmental, or practical impact.

The paper's main contribution is a controlled comparison of CNN-based and Vision Transformer-based visual encoders within a shared Transformer-decoder framework. The comparison of caption quality, model size, training efficiency, and inference latency provides useful empirical evidence regarding the trade-offs between the two encoder architectures. The finding that ResNet-50 achieves better results on several caption evaluation metrics while requiring substantially less training time is potentially useful for selecting models under computational constraints.

However, the novelty is primarily comparative and experimental rather than architectural. The authors should distinguish the contribution more clearly from existing image-captioning studies and explain what new insight the controlled experimental design provides. The societal and practical impact could be strengthened by discussing applications such as assistive technologies for visually impaired users, image retrieval, and automated content description. Evaluation of factual correctness, object hallucination, and accessibility-related caption quality would also help establish the broader value of the research. Overall, the work has practical potential, but its originality and wider impact require clearer justification.

5. Does the proposed work have practical relevance and real-world applicability? Provide your comments

The proposed comparison has practical relevance for selecting image-captioning models in applications such as assistive technologies, image indexing, digital content management, and human-computer interaction. The reported results indicate that ResNet-50 offers a favorable balance between caption quality, model size, and training time under the experimental conditions used. The similar inference latencies of the two systems also provide useful information for deployment planning.

Nevertheless, the evaluation is limited to the Flickr30k benchmark and a single reported experimental run for each model. Practical applicability would be better established through testing on additional datasets, diverse real-world images, and more challenging scenes. The authors should examine caption factual accuracy, object hallucination, and the ability to describe relationships between multiple objects. Memory consumption, inference hardware requirements, and performance under constrained computing conditions should also be reported where possible. Human evaluation of caption relevance, fluency, and correctness would complement the automatic metrics. The findings are promising, but broader validation is needed before generalizing the conclusions to real-world deployment.

6. References: Ensure the paper includes at least 10–15 relevant references, preferably published between 2021 and 2026, and check whether all references are appropriately cited within the manuscript. (Provide comments)

The manuscript contains 17 references covering image captioning, CNN architectures, Vision Transformers, Transformer decoders, multimodal learning, and automatic caption evaluation metrics. The inclusion of recent survey papers from 2025 and 2026,

along with foundational publications, provides a reasonable starting point for the literature review.

However, the reference list should be expanded with approximately 10–15 additional relevant and recent publications from 2021–2026, where available, particularly on CNN–Transformer comparisons, modern image-captioning architectures, vision-language pretraining, factual consistency, hallucination evaluation, and human-centered caption assessment. The literature review should more explicitly compare the proposed experimental setup and findings with closely related studies. All in-text citations should be checked against the reference list, and bibliographic details, publication venues, years, and DOI information should be verified. The code repository URL should also be checked for accessibility and consistency with the implementation described in the paper. These revisions would improve the literature coverage and establish the study's contribution more convincingly.

7. Overall comments

The manuscript presents a useful experimental comparison of ResNet-50 and ViT-B/16 encoders for image caption generation using a common Transformer decoder. Its strengths include a clearly stated research objective, a reasonably detailed methodology, multiple standard caption-quality metrics, and an assessment of model size, training time, throughput, and inference latency. The reported results suggest that ResNet-50 provides a favorable quality-efficiency trade-off on the Flickr30k dataset, while ViT-B/16 performs slightly better on BLEU-1 to BLEU-3.

However, the contribution is primarily comparative, and the manuscript would benefit from a stronger demonstration of novelty, more extensive experimental validation, and a deeper analysis of the relatively small performance differences. Multiple training runs, statistical significance testing, human evaluation, and assessment of factual correctness would improve confidence in the conclusions. The authors should also clarify the handling of spatial visual features, provide complete reproducibility details, and strengthen the discussion of generalization beyond Flickr30k. The reference list and comparison with recent image-captioning research should be expanded. Overall, the paper addresses a relevant research problem and provides useful preliminary findings, but further validation and clarification are needed to strengthen its scientific contribution and practical significance.

Reviewer #3

Questions

1. Is the paper relevant to the scope of the conference?

yes

2. Technical quality: Provide comments based on the methodology, mathematical justification for proposed work, architecture/block diagram of proposed method, and experimental validation

Technical Quality – Reviewer Comments:

The paper presents a systematic comparison of ResNet-50 and ViT-B/16 encoders using a shared Transformer decoder for image caption generation. The methodology is reasonably described, with mathematical and architectural details provided for both models. However, the justification for the selected architectures and the impact of encoder representations on caption quality could be strengthened. Although quantitative results are reported using standard evaluation metrics, additional experiments, statistical significance analysis, and comparisons with recent state-of-the-art methods are needed to establish the robustness of the findings.

4. Novelty and Societal Impact: Provide comments on the originality of the proposed work and its potential societal, industrial, environmental, or practical impact.

Novelty and Societal Impact – Reviewer Comments:

The paper presents a useful comparative study of CNN-based and Vision Transformer-based encoders for image caption generation using a common Transformer decoder. Its originality lies mainly in the controlled comparison of the two encoder architectures and their computational trade-offs rather than in proposing a new algorithm. The work has potential applications in assistive technologies for visually impaired individuals, image retrieval, and automated image description. However, the novelty could be strengthened by comparing the models with recent advanced captioning methods and evaluating their practical effectiveness in real-world scenarios.

5. Does the proposed work have practical relevance and real-world applicability? Provide your comments

Practical Relevance and Real-World Applicability – Reviewer Comments:

The proposed work has practical relevance in automated image description, assistive technologies for visually impaired individuals, and image retrieval systems. The comparison of ResNet-50 and ViT-B/16 provides useful insights into the trade-off between caption quality and computational efficiency. However, validation on diverse real-world images and comparison with advanced captioning models are needed to establish broader applicability.

6. References: Ensure the paper includes at least 10–15 relevant references, preferably published between 2021 and 2026, and check whether all references are appropriately cited within the manuscript. (Provide comments)

References – Reviewer Comments:

The manuscript includes 17 references covering image captioning, CNNs, Vision Transformers, and evaluation metrics. However, several references are older foundational works. The authors should incorporate more recent studies published between 2021 and 2026 on Transformer-based image captioning and vision-language models. All references should be verified for completeness, relevance, and consistency

with in-text citations. The literature review should also better establish the research gap in comparison with recent state-of-the-art approaches.

7. Overall comments

Overall Comments – Reviewer Evaluation:

The paper presents a relevant comparative study of CNN-based and Vision Transformer-based encoders for image caption generation. The methodology is reasonably structured, and the experimental results provide useful insights into caption quality and computational efficiency. However, the originality of the work is primarily comparative, and the evaluation would benefit from additional comparisons with recent state-of-the-art methods and statistical significance analysis. The authors should also strengthen the discussion of practical applications, incorporate recent literature, and improve the analysis of the experimental findings. Overall, the paper has potential, but some improvements are required to strengthen its scientific contribution and practical relevance.

Reviewer #4

Questions

1. Is the paper relevant to the scope of the conference?

Yes. A controlled CNN vs ViT encoder comparison for image captioning fits the conference's AI/vision scope.

2. Technical quality: Provide comments based on the methodology, mathematical justification for proposed work, architecture/block diagram of proposed method, and experimental validation

The paper compares ResNet-50 and ViT-B/16 encoders paired with an identical two-layer Transformer decoder on Flickr30k (Karpathy split). The implementation details are clearly reported: vocabulary, preprocessing, Optuna search, learning rates, seed, timing protocol and code repository. Reporting at this level is commendable. The reported parameter counts, model sizes and timing figures are internally consistent. The following points should be addressed.

Single-vector encoder memory. Both encoders are reduced to one 512-d vector (pooled ResNet features; ViT CLS token), which is the only memory the decoder cross-attends to. Cross-attention over one token is trivial, and neither ResNet's spatial grid (7×7 = 49 regions) nor ViT's 196 patch tokens reach the decoder. The comparison therefore measures the quality of two global image embeddings, not "CNN local feature extraction vs ViT global contextual modelling" as claimed in the Introduction and Discussion. Please either (a) feed the spatial or patch token sequences to the decoder, or (b) clearly restate the scope and soften the conclusions. Section III.A also describes "the spatial feature map from the penultimate global average pooling layer", which is self-contradictory: after global average pooling there is no spatial map.

Statistical support for the conclusion. The quality differences are very small (BLEU-4

+0.0029, METEOR +0.0009, ROUGE-L +0.0037, CIDEr +0.0126) and come from a single run with one seed. Run-to-run variation on Flickr30k is typically of this size or larger. Please report the mean ± standard deviation over at least 3 seeds, or bootstrap confidence intervals over the 1,000 test images. Otherwise, state that caption quality is comparable and base the recommendation on efficiency, which shows clear differences.

Table I: the encoder backbone counts (25,557,032 and 86,567,656) include the ImageNet classification heads (about 2.05 M and 0.77 M parameters). The text says these heads were removed, and the reported file sizes are consistent with their removal. Please correct the counts (ResNet-50 without fc: 23,508,032; ViT-B/16 without head: 85,798,656) and the totals.

Please state which tool computed the metrics (pycocoevalcap or NLTK) and the tokenisation used. METEOR and CIDEr values differ noticeably between implementations.

The hyperparameter search ran 15 trials of only 2 epochs each, while the final models ran 24 epochs. ViT received a very low encoder learning rate (3.3×10^{-6}) and high dropout (0.32). Please discuss whether this short-horizon search may have disadvantaged ViT, given that the paper's main finding is a ViT vs ResNet comparison.

Please specify Adam or AdamW for each model ("Adam/AdamW" is ambiguous) and the decoder feed-forward dimension.

Section VI.D describes qualitative differences on crowded scenes, but no examples are shown. Please add a figure with 3–4 test images and both models' captions, including failure cases.

Please add an architecture block diagram and training/validation loss curves for both models, since convergence behaviour is discussed in the text.

Flickr30k contains 31,783 images; the Karpathy split uses 31,014 of them.

The effective throughput uses the total epoch time, which includes validation and data loading. Please note this in the table, or report training-only throughput.

4. Novelty and Societal Impact: Provide comments on the originality of the proposed work and its potential societal, industrial, environmental, or practical impact.

The work is a controlled empirical comparison rather than a new method. That is acceptable, and useful, for a conference paper, since such matched comparisons are less common than new-architecture papers. Its value would increase substantially if the decoder attended to spatial or patch tokens (see Technical Quality, item 1), because that is the setting in which the CNN vs ViT question matters most. The findings are relevant to practitioners choosing encoders under limited compute budgets, for example in assistive captioning applications for visually impaired users.

5. Does the proposed work have practical relevance and real-world applicability? Provide your comments

The efficiency analysis has clear practical value: ViT trains 2.31× longer and produces a 2.75× larger model, while inference latency is similar. This gives concrete guidance

for training on limited GPUs such as the Kaggle T4. The public code repository and the detailed reproducibility information are strong points.

6. References: Ensure the paper includes at least 10–15 relevant references, preferably published between 2021 and 2026, and check whether all references are appropriately cited within the manuscript. (Provide comments)

The paper has 17 references, all cited in the text, and seven are from 2021–2026 ([1], [2], [8]–[12]). Suggestions:

References [1] (Al-Malla et al., Journal of Big Data, 2026, “in press”) and [2] (Abdulgalil and Basir, NLP Journal, vol. 12, 2025) could not be located. Please verify them and add DOIs.

The claim that “relatively few studies” perform controlled encoder comparisons should be supported by citing related recent work, for example CPTR (Liu et al., 2021, full Transformer captioning with a ViT encoder), ClipCap (Mokady et al., 2021), and studies on grid vs region features for captioning (Jiang et al., CVPR 2020).

Please cite the metric implementation used (e.g., the COCO caption evaluation toolkit) and Optuna (Akiba et al., KDD 2019).

Please add venue details (volume and pages) for the conference references to keep the formatting consistent.

7. Overall comments

Recommendation: Accept with modifications

This is a clearly written, well-documented comparative study with unusually thorough reproducibility details: hyperparameters, seeds, timing protocol and code. The reported parameter counts, model sizes and timing figures are internally consistent. The efficiency comparison is convincing and practically useful.

The main weaknesses are:

(1) Both encoders are reduced to a single global vector, so the study does not actually compare CNN local features with ViT global attention as claimed.

(2) The caption-quality differences are very small and come from a single run, so the conclusion that ResNet-50 gives better quality is not statistically supported.

(3) The Table I parameter counts include the removed classification heads.

(4) There are no figures: no architecture diagram, loss curves or qualitative examples.

The authors should:

restate the scope, or use spatial/patch tokens as decoder memory;
report multi-seed or bootstrap variability and soften the quality claim to “comparable”;
correct Table I;
specify the metric toolkit;

add the architecture diagram, loss curves and qualitative caption examples; verify references [1] and [2] and add the suggested recent related work.